Phenological sensitivity to climate change disrupts species' coexistence

D.M. Buonaiuto [1,2,3,a], M. Donahue[4], E.M. Wolkovich[2,3,5]

May 1, 2026

Author affiliations:

[1]Department of Plant Science and Landscape Architecture , University of Maryland, College Park, Maryland, USA. ORCID: 0000-0003-4022-2591

[2]Arnold Arboretum of Harvard University, Boston, Massachusetts, USA.

[3]Department of Organismic and Evolutionary Biology, Harvard University, Cambridge, Massachusetts, USA

[4]Hawai'i Institute of Marine Biology, University of Hawai'i at Manoa, Kan'eohe, HI, USA.

[5]Forest & Conservation Sciences, Faculty of Forestry, University of British Columbia, Vancouver, British Columbia, Canada

[a]Corresponding author: 617.823.0687; dbuona@umd.edu

1 Abstract

Differences in phenology, the seasonal timing of life-cycle events such as hatching, leafout and germination, are recognized as an increasingly important component of species' coexistence. Observed shifts in phenology due to climate change are likely to restructure phenologically-mediated species interactions, but current coexistence frameworks are not equipped to model this relationship in a changing environment. Here, we re-examine phenology as the outcome of an interaction between the environment and each species' unique physiology, genetic and developmental attributes, which in turn impacts competition. We argue that this flexible and simple—yet more biologically-accurate approach—could revolutionize forecasting of community assembly with climate change. Drawing on examples from empirical germination studies and common models from modern coexistence theory we outline new approaches to bridge across intra- and inter-annual time, and review how phenological-mediated patterns of coexistence under a historical climate regime may degrade to competitive exclusion with climate change. Importantly, the biological realism of our framework provides a major step forward towards translating ecological theory into practice and advancing predictive ecology.

2 Introduction

Each spring, in ecosystems across the temperate zone, ecological communities re-assemble from their period of winter dormancy with the hatching of insects, arrival of migratory birds, and emergence of new seedlings, leaves and flowers. The seasonal timing of these life cycle events—their phenology—can be quite different, even among closely related organisms in a shared habitat (1; 2). These phenological differences among taxa have often been overlooked in ecological theory outside of plant-pollinator interactions, but growing evidence suggests that they may play a major role in dictating community interactions and shaping the structure and function of ecological communities (3; 4; 5).

Experiments show that the order in which species begin their seasonal growth can alter the outcomes of competitive interactions (6; 7). Such phenological differences among species can generate seasonal priority effects where earlier active species gain a competitive advantage (8; 9). This advantage may occur through drawdown of limiting resources before other competitors emerge or arrive, and other ways that modify the environment to the detriment of later emerging species (10; 11). Studies in plants (12; 13), amphibians (14), insects (15; 16) and birds (17) have found that apparently small phenological priority effects, on the order of just a few days, can substantially alter competitive hierarchies in ways that persist across several growing seasons (18).

The importance of seasonal priority effects in mediating interspecific competition suggests how phenology may be fundamental to coexistence (19; 4), but not how phenological differences between species arise. Functionally, phenology is the realized product of an interaction between the environment (cues) and a species' unique physiology, genetic and developmental attributes (20). Differences in how species respond to the same environmental cues are one of the primary mediators of the trade-off between phenology and competition. With environmental cues changing rapidly in response to anthropogenic warming, phenological shifts have become one of the most pronounced indicators of climate change to date. (21). Understanding the degree to which a species responds to environmental variation, known as phenological sensitivity (Δ phenology/ Δ environmental cue), is key to predicting community assembly in an era of global change. Below we show why integrating phenological sensitivity into community assembly is critical, and outline a path to predict assembly.

3 How phenological differences arise

Environmental cues such as temperature, light and moisture are the primary triggers that control phenology (22), with many species responding to multiple aspects of cues (23; 24). For example, many temperate plants that rely on temperature as a cue require exposure to both cool winter (0-4°C, hereafter: winter chilling, which is similar to cold stratification) followed by warm spring temperatures to initiate phenological transitions (25). Controlled experiments have helped quantify species-specific responses to cues, which—when combined with the exact environmental conditions—drive differences in realized phenology between species.

Decades of seed germination studies provide reliable examples of how realized phenology can vary across species × environment (e.g., 26). We illustrate this by re-analyzing seed phenology data from (27) using accelerated failure time models (28) (see Supporting Information for details), in which 11 species from one habitat were germinated under varying temperature conditions to determine their sensitivities. These co-occurring species showed very different sensitives to the same environmental cues (Figure 1a) that could potentially impact competition. For example, for *Persicaria virginiana* and *Eurybia divaricata*—a pair of woodland herbs that frequently compete in Eastern Temperate Forests—which species germinates first depends on environmental conditions, and the order switches with increasing cue levels (see Figure 1b,c; in this case, chilling duration acts as the cue). This illustrates that the strength of the seasonal priority effect is highly dependent on the environment, yet this is not how most coexistence models approach phenology.

4 Recent work integrating phenology into assembly models

Growing interest in how phenology may structure communities has led to new assembly models that center on the relationship between timing and competition (5; 8; 9). Building off experimental (29) and observational (30) evidence that earlier species appear to perform better, models have generally included a trade-off between timing and competition. This trade-off has been expressed explicitly through a connection of phenological differences to parameters controlling the strength of competition, or implicitly by providing a fitness advantage to earlier species offset by poorer performance under low resource conditions (e.g., R^* in 8; 9).

Models providing a fitness advantage to earlier species have all recognized the importance of environmental variability in communities structured in part by phenological differences, generally including interannual variation in the environment that in turn drives differences in phenology. Many models—focused often on grasslands and water competition—have assigned phenology as equivalent to an intrinsic species-level timing (or season length strategy). Year to year variation in the environment then alters absolute differences between species, which then consistently co-varies with resource competition (29; 9). In these approaches the relative orders between species remain fixed, but other models allow species relative orders and timings to vary year-to-year assuming the underlying relationship between pairwise differences in timings and interspecific-competition is known—and static (5). Each of these approaches thus fixes one aspect of what is actually a dynamic environment × phenology × competition relationship (i.e., either through an explicit fixed relationship between phenology and competition or fixing the order of species but allowing the strength of competition to vary). This is an understandable simplification, but one that may make forecasting community shifts with climate change challenging, as the environment becomes more variable.

Phenological models of community assembly have included environmental variability (5; 9) that assumes an underlying stationary environment (one that varies year-to-year but with an underlying constant distribution), in contrast to the reality of climate change. Temperature, for example, trends higher in its mean over time (with potential changes to the variance, 21)—creating a non-stationarity environment. Because environmental variability in assembly models structures what species combinations are stable, shifts in the environment with anthropogenic climate change are likely to reshape communities, but forecasting them requires accurately capturing how the environment affects phenology and competition together in a simple—but still robust—form. This means capturing how the environment can determine both the absolute (e.g., earlier in the year) and relative (e.g., order of arrival) timings of species, and the how these timings impact coexistence. While current models to date have simplified one aspect of this problem, it may be possible to address it through new approaches.

5 Climate Change, Phenology and Coexistence

Leveraging community assembly models to address how shifting phenology with climate change may affect coexistence requires models that work across timescales. While many coexistence models include inter-annual environmental variability, including it intra-annually as well would allow modeling phenology more physiologically, which would better capture how the competition depends on the environment filtered through species phenological sensitivities. This modeling approach may also show how climate change can alter the stability of trait-based, community trade-offs.

If community assembly depends on species timings, the shifts in environmental cues with climate change could undermine coexistence. To illustrate this possibility, we extended a traditional assembly model to include phenological sensitivity in its simplest form—two species with differential sensitivity to one environmental cue. Our model was similar to others in varying species intrinsic competitive abilities at the same time we varied phenology, but differed in modeling phenological sensitivities alongside intra-annual environmental variation. Any changes in the environment thus filter through species-specific sensitivities, and transform differences in realized phenology between species (Figure 2b). Comparing long-term co-occurrence (which we consider coexistence here) of species pairs under two different climate regimes, we found that no pair coexisting under one climate could coexist under the other (Figure 2a). Thus, for any species that currently coexist due to the tradeoff between relative phenology and intrinsic competitive ability, climate change disrupts this coexistence mechanism.

Improving the predictive power of coexistence models with climate change also likely requires rethinking how we measure species phenology. Our example model showed that the required tradeoff between differences in realized phenology and competitive differences is stable across a shifting environment (i.e., the same magnitude of phenological differences are required to offset the same level of differences in competitive ability), and thus uninformative on its own for predicting which species pairs can coexist. Instead, it is the tradeoff between phenological sensitivity differences and competition differences that change in a novel environment (Figure 2a,c). Thus, measuring species phenologies may be far less useful for understanding future community assembly than measuring their phenological sensitivities.

6 Paths Forward

Phenological sensitivity is a trait that can be readily measured for real species, providing a link between assembly theory and observed ecological changes. Yet, effectively leveraging this trade-off between phenological sensitivity and competitive ability to model community assembly under future climates will require major efforts by the modeling and phenology-focused communities. Useful predictions will need the continued development of both novel experimental approaches and more-biologically realistic modeling frameworks.

On the experimental side, more studies that quantify species-level phenological sensitivity could help answer open questions in ecophysiology with major implications for forecasting. Studies that estimate the phenological sensitivity to multiple environmental cues could address how much local adaptation versus plasticity structure phenological sensitivity and competitive traits (e.g. (31; 32), but will be most useful for forecasting if they include a greater scope of taxa and lifeforms. While much ecological work has focused on annual plants, herbaceous perennials and woody plants are the dominant functional types in many ecological systems. For these functional types, phenology-mediated competition occurs between individuals both within- (i.e., seedling vs. seedling or mature plant vs mature plant) and across- (i.e, mature plant vs. seedling) life stages. Studies in woody plants suggest separate phenological phases can respond quite differently to the same environmental cues (33) and that juveniles and adults express substantially different phenological patterns in the same environments (34), yet changes in phenological sensitivity to multiple cues across life stages (i.e., seed germination vs vegetative cues) have not been widely explored.

On the modeling side, researchers should integrate the environmental drivers of more complex cues to assembly models. Current models—including those conceptualized for seed germination—assume phenology is independent of (or weakly linked to) the environment, but research has repeatedly shown than phenology is determined to multiple, interacting environmental cues. For seed germination this often includes multiple aspects of temperature (i.e., forcing and chilling), light (daylength or irradiance) and moisture (35; 22). Exactly how best to include these cues and their drivers should likely be determined by the main factor that selects for them—environmental risk. For example, leafout of many forest trees relies on cues of winter chilling and/or photoperiod, which could reduce the risk of frost damage in unstable spring environments. Introducing biologically realistic mortality events into community assembly models that represent important selective forces, such as late-season frosts or early droughts, offers a straightforward approach to assess the importance of the often overlooked aspect of phenology. Ecological models generally subsume such risks into species traits (e.g., increasing maintenance costs), which misses the reality that such risks are also product of timing $\times$ environment.

A changing risk landscape associated with a changing environment may be the underlying driver of observed correlations between the timing and intensity component of life cycle transitions (i.e., germination phenology and total germination percent of a seed cohort, 35). But models will need to explicitly include the environment and phenology to predict this. Further, for plants—and likely other organisms—the timing and fraction of phenological success (e.g., seed germination or leafout) usually covary in nature, which likely has important implications for coexistence. Building models that can isolate these two trade-off axes of coexistence—timing and abundance (fraction)—would allow ways to assess the relative importance of these within- and across season trade-offs.

Community assembly modeling could have the flexibility to incorporate many of the ecological aspects we discussed above (multiple cues, environmental risk, within- and across- season bet-hedging). Meaningful predictions will require models that capture underlying physiological processes, and experiments that can readily inform model parameters and development. Such approaches would translate ecological theory into practice and advance a more predictive ecology than is better equipped to inform ecosystem conservation and management.

Figures

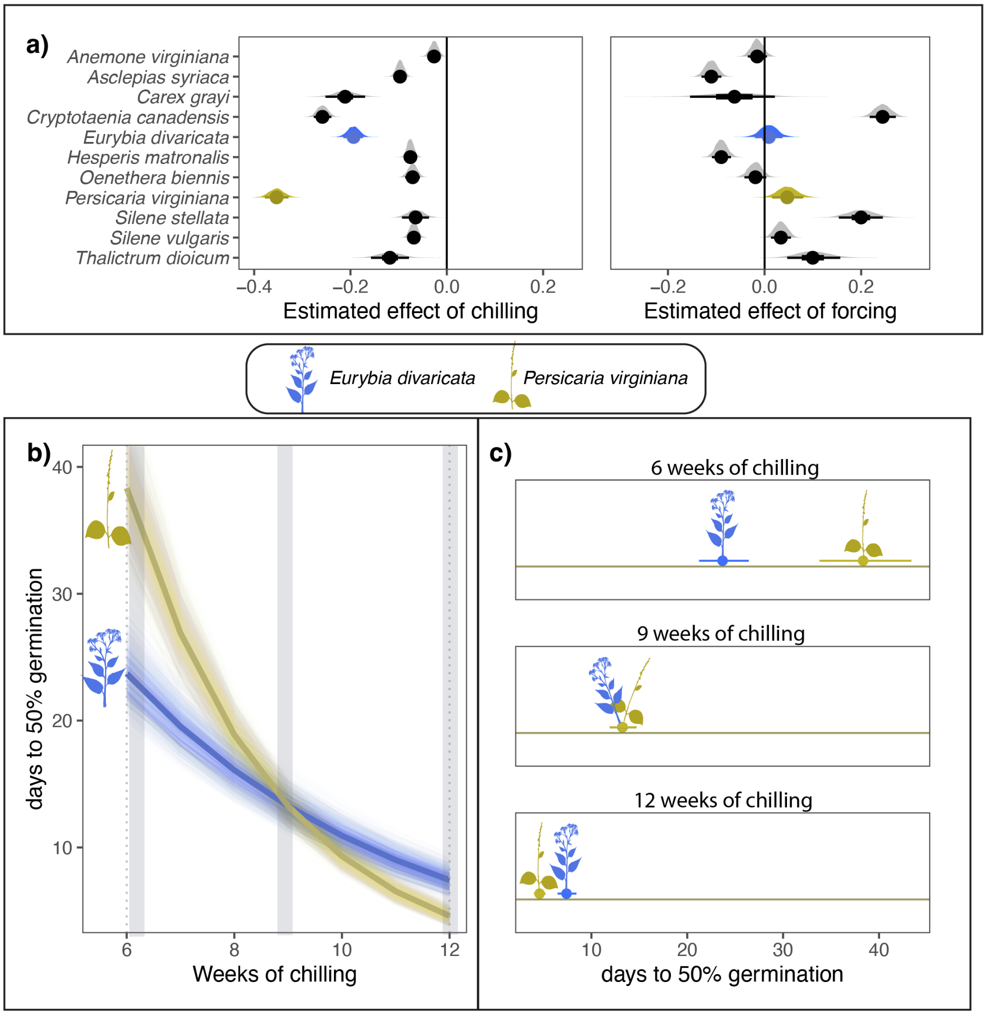

Figure 1: **Co-occurring species can differ substantially in their phenological sensitivity (Δ phenological event/Δ climate) to the environment.** Panel a) estimates the phenological sensitivity of germination timing to winter chilling and spring forcing for eleven herbaceous species of eastern North America. Points represent the mean estimate and thick and thin bars represent the 50 and 95% uncertainty intervals (with posteriors shown above lines as gray distributions; for details on the model, see the Supporting Information). In b) and c) we highlight the phenological sensitivity to winter chilling of two of these species—*Eurybia divaricata* and *Persicaria virginiana*—that are commonly observed in competition. b) Depicts the sensitivity of each species to winter chilling, with thick lines representing the mean estimate and lighter lines representing modeled uncertainty. Panel c) illustrations how the species' different sensitivities manifest in contrasting realized germination differences under different climate conditions.

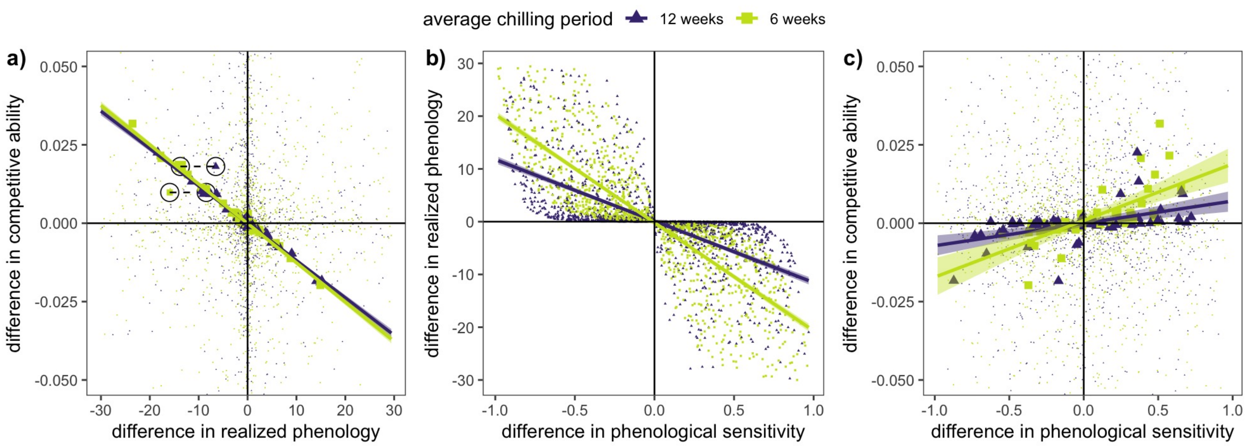

Figure 2: **Climate change will alter the relationships between competitive ability, phenological sensitivity, realized phenology and coexistence.** Panel a) depicts the trade-off between differences in realized phenology and differences in competitive abilities (in all panels each point represents a species pair with randomly selected phenological sensitivities and competitive abilities subjected to two climate scenarios—12 and 6 weeks of chilling; species pairs that co-occurred after 300 years are in bold). The trade-off needed to maintain coexistence does not change between climate scenarios, but no species pairs that coexisted under one scenario can coexist under the other scenario, as highlighted by the two sets of species pairs circled in panel a). Panel b) shows the relationship between differences in phenological sensitivity and differences in realized phenology, which depend on the environment. Panel c) depicts the trade-off between differences in phenological sensitivity and differences in competitive ability under the two climate scenarios (see Supporting Information for details).

References

[1] Ovaskainen, O., Skorokhodova, S., Yakovleva, M., Sukhov, A., Kutenkov, A., Kutenkova, N., Shcherbakov, A., Meyke, E., and del Mar Delgado, M. *Proceedings of the National Academy of Sciences* **110**(33), 13434–13439 (2013).

[2] An, S., Chen, X., Shen, M., Zhang, X., Lang, W., and Liu, G. *Frontiers in Plant Science* **Volume 13 - 2022** (2022).

[3] Cleland, E. E. and Wolkovich, E. *Annual Review of Ecology, Evolution, and Systematics* **55**(Volume 55, 2024), 471–492 (2024).

[4] Blackford, C., Germain, R. M., and Gilbert, B. *The American Naturalist* **195**(6), E168–E180 (2020). PMID: 32469665.

[5] Rudolf, V. H. W. *Ecology Letters* (2019).

[6] Dawson-Glass, E., Schiafo, R., Kuebbing, S. E., and Stuble, K. L. *Ecology* **106**(1), e4504 2025/07/15 (2025).

[7] Buonaiuto, D. M. and Wolkovich, E. M. *Journal of Ecology* **111**, 1711–1721 2023/07/19 (2023).

[8] Wolkovich, E. and Donahue, M. J. *Biological Reviews* **96**(6), 2810–2827 (2021).

[9] Levine, J. I., Levine, J. M., Gibbs, T., and Pacala, S. W. *Ecology Letters* **25**(5), 1110–1125 (2022).

[10] Fukami, T. *Annual Review of Ecology, Evolution, and Systematics* **46**(1), 1–23 (2015).

[11] Weidlich, E. W. A., Nelson, C. R., Maron, J. L., Callaway, R. M., Delory, B. M., and Temperton, V. M. *Restoration Ecology* **n/a**(n/a), e13317 2020/12/17 (2020).

[12] Wainwright, C. E., Wolkovich, E. M., and Cleland, E. E. *Journal of Applied Ecology* **49**(1), 234–241 Nov (2011).

[13] Koerner, C., Stoecklin, J., Reuther-Thiebaud, L., and Pelaez-Riedl, S. *New Phytologist* **177**(3), 698–705 (2008).

[14] Rudolf, V. H. W. and Singh, M. *Oecologia* **173**(3), 1043–1052 (2013).

[15] Ekholm, A., Tack, A. J. M., Pulkkinen, P., and Roslin, T. *Journal of Animal Ecology* **89**(3), 829–841 (2020).

[16] Rasmussen, N. L., Van Allen, B. G., and Rudolf, V. H. W. *Journal of Animal Ecology* **83**(5), 1206–1215 (2014).

[17] Samplonius, J. M. and Both, C. *Current Biology* **29**(2), 327–331.e2 2025/07/15 (2019).

[18] Vaughn, K. J. and Young, T. P. *Ecological Applications* **25**(3), 791–799 2022/03/02/ (2015).

[19] Diez, J., Schlauch, J., DuCharme, E., Collings, J., and Erskine, S. *Journal of Ecology* **n/a**(n/a).

[20] Donohue, K., DORN, L., Griffith, C., Kim, E., Anna, A., Polisetty, C. R., and Johanna, S. *Evolution* **59**(4), 758–770 04 (2005).

[21] Canadell, J., Monteiro, P., Costa, M., Cotrim da Cunha, L., Cox, P., Eliseev, A., Henson, S., Ishii, M., Jaccard, S., Koven, C., Lohila, A., Patra, P., Piao, S., Rogelj, J., Syampungani, S., Zaehle, S., and Zickfeld, K. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.* Cambridge University Press, New York, NY, (2021).

[22] Forrest, J. and Miller-Rushing, A. J. *Philosophical Transactions of the Royal Society B: Biological Sciences* **365**(1555), 3101–3112 Sep (2010).

[23] Lang, W., Zhang, Y., Li, X., Meng, F., Liu, Q., Wang, K., Xu, H., Chen, A., Peñuelas, J., Janssens, I. A., and Piao, S. *Nat Ecol Evol* **9**(2), 261–272 Feb (2025).

[24] Chuine, I. and Régnière, J. *Annual Review of Ecology, Evolution, and Systematics* **48**(Volume 48, 2017), 159–182 (2017).

[25] Ettinger, A. K., Chamberlain, C. J., Morales-Castilla, I., Buonaiuto, D. M., Flynn, D. F. B., Savas, T., Samaha, J. A., and Wolkovich, E. M. *Nature Climate Change* **10**(12), 1137–1142 (2020).

[26] Fernández-Pascual, E., Carta, A., Mondoni, A., Cavieres, L. A., Rosbakh, S., Venn, S., Satyanti, A., Guja, L., Briceño, V. F., Vandelook, F., Mattana, E., Saatkamp, A., Bu, H., Sommerville, K., Poschlod, P., Liu, K., Nicotra, A., and Jiménez-Alfaro, B. *New Phytologist* **229**(6), 3573–3586 (2021).

[27] Buonaiuto, D. M. and Wolkovich, E. M. (2023).

[28] Onofri, A., Gresta, F., and Tei, F. *Weed Research* **50**(3), 187–198 2022/03/02 (2010).

[29] Godoy, O. and Levine, J. M. *Ecology* **95**(3), 726–736 (2014).

[30] Wolkovich, E. M. (2013).

[31] Lipowsky, A., Roscher, C., Schumacher, J., Michalski, S. G., Gubsch, M., Buchmann, N., Schulze, E.-D., and Schmid, B. *Perspectives in Plant Ecology, Evolution and Systematics* **17**(1), 66–77 (2015).

[32] Liang, L. and Wu, J. *International Journal of Biometeorology* **65**(11), 1953–1966 (2021).

[33] Buonaiuto, D. M. and Wolkovich, E. M. *Journal of Ecology* **109**(8), 2922–2933 2025/07/31 (2021).

[34] Vitasse, Y. *New Phytologist* **198**(1), 149–155 2025/07/31 (2013).

[35] Baskin, C. and Baskin, J. *Seeds: Ecology,Biogeography, and Evolution of Dormancy and Germination.* Elsevier Inc, (2014).